

Analog Electronics Homework 2

1. Analyze the full-wave rectifier circuit given in Figure 1. Sketch v_{out} and determine dc voltage available. Diodes are ideal.

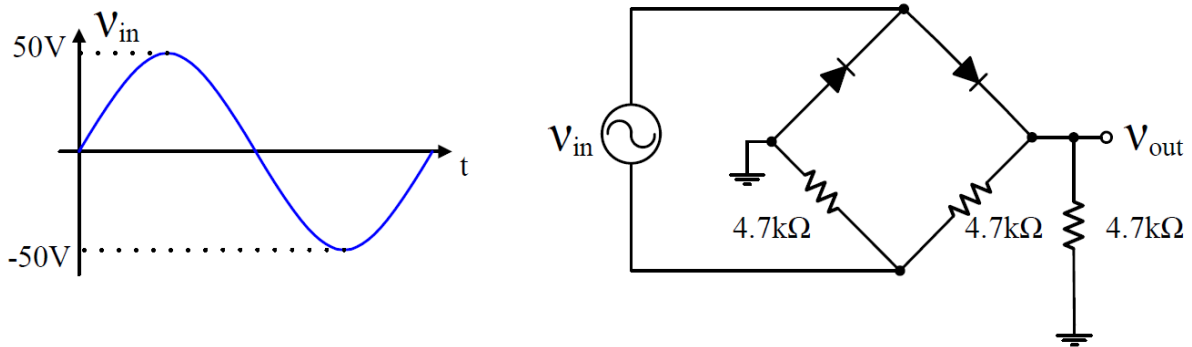


Figure 1

2. Analyze the clipper circuits given in Figure 2(a) and Figure 2(b). Sketch v_{out} for each part for the given sinusoidal input.

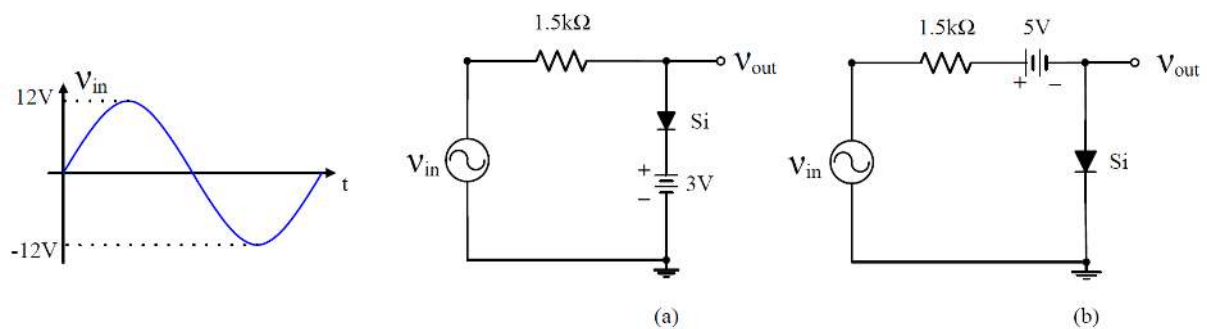


Figure 2

3. Analyze the clipper circuit given in Figure 3. Sketch v_{out} for the given triangular input.

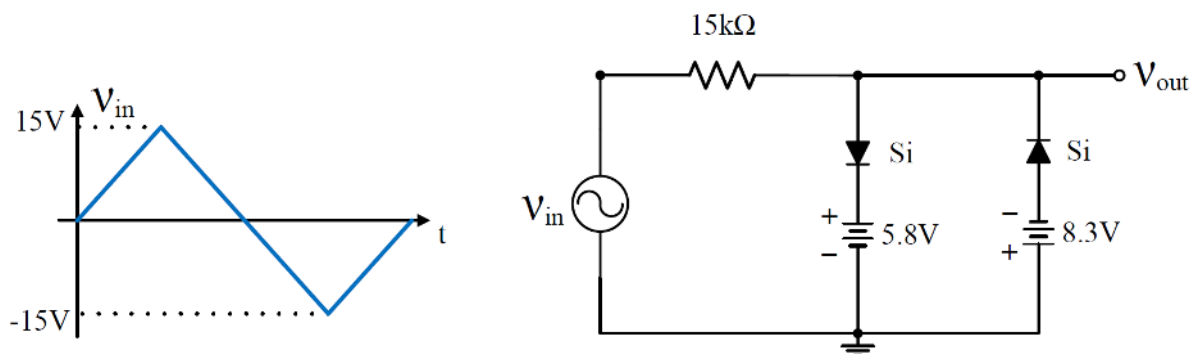


Figure 3

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4. Analyze the clamper circuits given in Figure 4(a) and Figure 4(b). Sketch v_{out} for each part for the given square input.

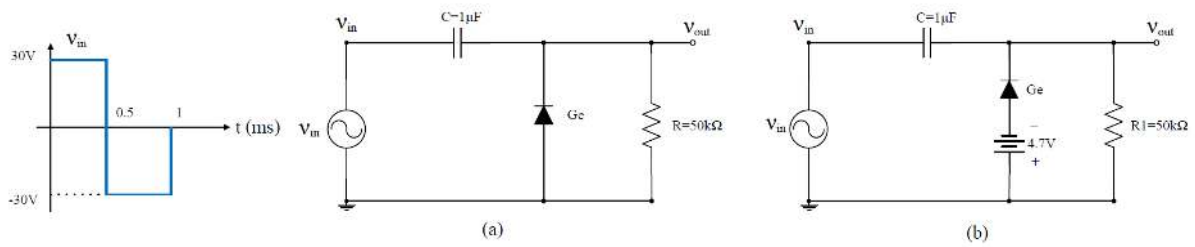


Figure 4

5. a) Analyze the Zener voltage regulator circuit given in Figure 5. Find V_L , I_L , I_Z and I_R if $R_L=100\Omega$.
 b) Repeat part (a) if $R_L=680\Omega$.
 c) Find R_L for the maximum power dissipation of Zener diode.
 d) What is the minimum R_L that will keep Zener diode in the “on” state.

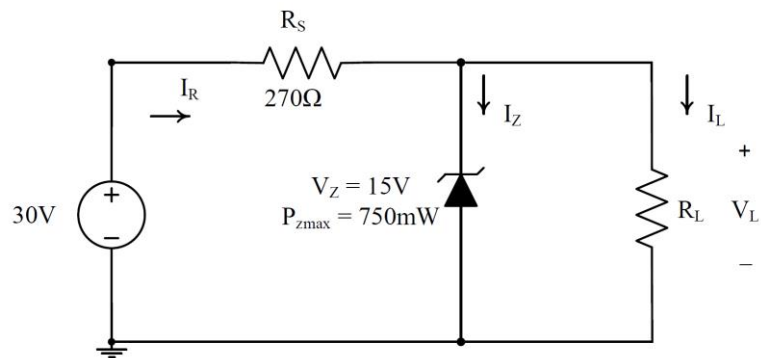


Figure 5